DTU Management
Department of Technology, Management and Economics

Competing for molecules
Network Optimization of Cross-Industry Allocation

Master Thesis

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Approval

This thesis has been prepared over six months at the Section for Indoor Climate, Department of Civil Engineering, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

Author - s123456

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Signature

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Date

Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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Contents

1 Introduction

The accelerating transition toward low-carbon energy systems has intensified research into the global production, transport, and allocation of green molecules such as green hydrogen, ammonia, e-methanol, and direct reduced iron (DRI). Existing modelling approaches span sector-specific optimisation, multi-carrier energy-trade modeling, and global commodity-flow assessments. However, these tools largely share a common characteristic: they assume cost-optimising behaviour, often embedded within political or economic conditions of individual countries, and therefore do not provide a neutral system-level view of how green molecules might distribute purely based on production availability and demand. The present study develops a model intended to fill this gap by exploring global flow patterns independent of national strategies, financing conditions, or trade preferences, focusing instead on the physical interaction between supply and demand under an open-market representation.

We are interested in understanding how green fuel is distributed between its production site and the various industries that purchase it for use. To illustrate this, we have chosen to focus on a small section of the world. This means that the conclusions presented in the report are not final, as not all production sites or all industries are represented. However, all conclusions in the report are made with the purpose of demonstrating that the model is correct and to show what it would be capable of with the correct input data.

1.1 Problem Statement

2 Theory

3 Literature review

3.1 Green Ammonia

3.1.1 Current landscape

Today, ammonia production is overwhelmingly fossil-based. It is produced by reacting nitrogen from the air with hydrogen derived from fossil fuels, typically natural gas or coal, through the Haber-Bosch process. Current global annual consumption is around 190 million tonnes, with the fertiliser industry accounting for approximately 70% of this demand. Ammonia synthesis is responsible for around 0.5 Gt of CO_2 annually, corresponding to roughly 1% of the greenhouse gas emmissions. [1].

Beyond fertilisers, ammonia has gained attention as a potential decarbonisation vector. Its high hydrogen content by weight and existing global trade infrastructure make it an attractive hydrogen carrier, enabling hydrogen produced in renewable-rich regions to be transported and reconverted elsewhere. This has brought ammonia into focus for new end-use sectors, for example maritime shipping and steel production, where direct electrification is technically difficult or economically unfeasible at scale. In addition, ammonia is being explored as a low-carbon fuel for power generation and long-distance transport, further expanding its potential role in the future energy system. [2]

As decarbonisation pressure intensifies, the share of ammonia produced from renewable electricity, so-called green ammonia, is expected to grow substantially. Demand is projected to nearly double by 2035, driven by both continued fertiliser demand and emerging energy applications such as maritime fuel and hydrogen transport [3].
In longer-term scenarios, growth could be even more pronounced: IRENA projects that global ammonia demand could reach around 688 million tonnes by 2050 in a 1.5°C-aligned pathway, more than three times current levels.

This highlights the importance of examining pathways for green ammonia in the context of increasing demand and uncertainty surrounding its future production.

3.1.2 Techno-economic considerations

As of today, green ammonia is not yet deployed at scale and remains an emerging technology, with significant implications for cost estimates. The price of fossil-based ammonia currently ranges between 110 and 340 USD per tonne, while production with carbon capture and storage increases costs to approximately 170 to 465 USD per tonne. Green ammonia production costs are significantly higher at 700 to 1400 USD per tonne depending on location and technology, but are expected to decline substantially, potentially reaching 300 - 500 USD per tonne by 2030 - 2050 in regions with favourable renewable resources [2].

This cost structure is dominated by the upstream hydrogen production step, which can account for up to 90% of total production costs [2]. The three principal cost drivers are therefore the price of renewable electricity, electrolyser capital costs, and system utilisation rates. In regions combining low-cost renewables with high capacity factors, green ammonia is expected to become increasingly cost-competitive over time as electrolyser technology matures and production scales up. However, the pace and geography of this cost decline are themselves highly uncertain, and techno-economic projections are sensitive to underlying assumptions about technology development and energy system conditions. The wide cost ranges reported in the literature reflect this uncertainty.

3.1.3 Demand

The demand for green ammonia is still uncertain and can be used in different applications. While some industries already rely on grey ammonia, such as the fertiliser industry, other industries could adopt green ammonia as a way of transitioning towards a more sustainable alternative.

Fertiliser industry

The fertiliser industry accounts for approximately 70% of global ammonia demand [1] and represents the most natural starting point for green ammonia adoption. Since ammonia is a direct and non-substitutable input to nitrogen-based fertilisers, demand in this sector is structurally high and relatively price-inelastic.

Steel industry

The steel industry is one of the hardest sectors to decarbonise and accounts for a significant share of global industrial CO_2 emissions. One emerging pathway is the use of hydrogen, potentially delivered via ammonia, as a reducing agent to replace coking coal in iron ore reduction.

In this case, ammonia acts as a hydrogen carrier: it is shipped to the production site and cracked back into hydrogen for use in the direct reduction process. The extent to which the steel sector will rely on green ammonia depends heavily on policy targets and the pace of technological transition.

Maritime shipping

Ammonia is one of the leading candidate fuels for decarbonising deep-sea maritime shipping, alongside methanol and hydrogen. Its existing global trade infrastructure and high energy density make it logistically attractive, and several major shipping companies are actively developing ammonia-fuelled vessels [3].

Unlike the fertiliser sector, shipping demand for ammonia is largely a function of regulatory drivers such as carbon pricing and emissions targets rather than a direct process requirement. As of now the absence of strong CO_2 pricing in shipping limits the near-term incentive to pay a premium for green fuels, though emerging regulatory frameworks could change this.

3.2 Challenges and Uncertainties

Despite the promising cost trajectory outlined above, several structural barriers must be overcome before large-scale green ammonia deployment becomes feasible.

On the supply side, realising projected cost reductions depends on a substantial build-out of electrolyser manufacturing capacity that does not yet exist. Supply chain bottlenecks, could delay the learning curve that underpins optimistic cost forecasts. Suitable production sites are also geographically concentrated in regions with abundant wind or solar resources, many of which are remote from existing industrial demand centres, adding logistical complexity to supply chain development.

On the demand side, green ammonia faces a classic coordination problem. Without sufficient and stable demand, producers cannot justify the capital expenditure required to build large-scale facilities. Without large-scale production, prices remain too high to attract widespread industrial adoption. Early off-take agreements and policy support are therefore critical to breaking this deadlock [3]. This is compounded by the fact that many potential end-users, particularly in fertiliser production, currently rely on cheap grey ammonia, and face limited near-term incentive to pay a green premium without regulatory pressure or contractual certainty.

Infrastructure presents a further challenge. While ammonia benefits from existing global shipping and port infrastructure, significant investment is still required in dedicated storage, bunkering facilities, and onshore distribution networks [4]. The toxicity of ammonia also imposes safety and regulatory constraints that add complexity and cost to its handling and transport, particularly in densely populated regions.

Finally, the policy environment remains deeply uncertain, as illustrated by recent developments in maritime shipping regulation. While the IMO adopted a revised GHG strategy in 2023 and approved a Net-Zero Framework in April 2025, formal adoption of the framework was postponed for one year in October 2025 following opposition from the United States, Saudi Arabia, and several major flag states [5]. The earliest possible entry into force is now 2029, introducing renewed uncertainty for fuel producers and investors. In the interim, the EU's FuelEU Maritime regulation remains the most substantive binding framework, but applies only within EU waters [6]. This regulatory fragmentation is itself a barrier to the green ammonia transition, as long-term investment in alternative fuel infrastructure requires policy certainty that is currently absent at the global level.

3.3 Previous work on the subject

Several optimisation models have been developed to analyse the supply and distribution of green ammonia and related hydrogen derivatives, though each differs in scope, methodology, and the questions it is designed to answer.

In the paper, *Optimal fuel supply of green ammonia to decarbonise global shipping* they present a cost minimisation model for supplying green ammonia to meet a proposed global shipping fuel demand in 2050. The model optimises infrastructure location, transport routes, and investment requirements across a global network of production and demand nodes, with the levelised cost of ammonia (LCOA) serving as the key metric for evaluating and comparing supply pathways.[7]

A strength in this model is the treatment of shipping-specific infrastructure and gives a clear indication of the large production facilities that are required for the large demand in the maritime sector to be fulfilled.

However, the model is specific to the maritime sector and does not consider competition for green ammonia across multiple industries.

The IRENA global hydrogen trade modelling effort [8] takes a similar perspective, exploring how hydrogen and hydrogen-derived commodities, could be produced and traded internationally by 2050. Its focus is on regional production potential, transport infrastructure costs, and the economics of different trade corridors. Like the model in, [7] it is primarily a planning tool and does not address how scarce supply would be allocated across competing demand sectors.

Our model differs from these in several important aspects. Rather than a optimal infrastructure planning model, it models the allocation of green molecules to different industries simultaneously. This allows us to examine how different demand nodes compete for available supply, which production sites are utilised, and where unmet demand concentrates geographically. The two approaches are complementary, infrastructure solutions derived from models such as [7] could serve as direct input to our framework, allowing us to evaluate how optimised supply chains perform when subjected to multi-industry competition across different scenarios.

Furthermore the model is designed as a flexible analytical tool where different future cases can be used as input. Due to the structure of the model industries or green molecules can be introduced without restructuring the underlying framework, and policies, prices, and demand levels can be adjusted to reflect different assumptions about adoption rates, production capacity, or regulatory environments. This makes the model well-suited to examine a wide range of conditions, from scenarios of supply scarcity, where competition between industries determines who receives green ammonia and where unmet demand concentrates, to scenarios of over production, where the question shifts to which regions and production sites are most utilised and which are competed out by cheaper alternatives.

4 Modeling

The problem is conceptualised as a network model in which the system is represented by nodes connected through edges that carry flow.

Sets:

To formulate the problem we start by defining a set for each type of node in the network. The network consists of production sites, transit points and finally off take centers, all connected through edges that define the transportation routes between the nodes. Let P denote the set of production sites, T denote the set of transition nodes, O the set of off take centers and N the full set of nodes in the network. All possible edges between all nodes are made and a cost matrix will define the cost for all edges (i, j) from node i to node j.

Parameters:

Each edge (i, j) is associated with a cost that refers to the cost of the transportation of green ammonia on the specific edge. We assume that the cost for each industry is the same hence no industry will pay more or less for the green ammonia that is being produced in each off take center. This makes the cost matrix only dependent on the nodes i and j. Edges between two transit nodes $t \in T$ are shipping routes, where edges between production sites $p \in P$ and transit nodes or from transit node to off take centers nodes $o \in O$ is routes onshore. The transportation cost is computed based on attributes such as the shipping rate, the geographical region of each node, the transportation mode used on that edge, the physical distance between nodes, and an estimate of the operating cost of onshore transportation. If transportation between two nodes are not possible the transportation cost will be stet to ∞. The cost matrix $c_{i,j}$ will therefore be a $N \times N$ matrix.

For each production site $p \in P$, the parameter $maxP^p$ specifies the maximum production capacity available. Similarly, each off take center $o \in O$ has a known demand level D^o, which the network attempts to satisfy through delivered quantities or, if necessary, unmet demand penalized through a parameter π. Since the problem is a minimization problem, we set π to be a high cost. In this case we take the the highest production cost from the off take centers and add the highest transportation cost.

$$\pi = 360 + 90 = 450$$

This will ensure that the model will distribute the green ammonia produced at the production sites before meeting the rest of the demand with the high cost. This is mostly effective in a scenario where there is less green ammonia produced than demand which is how it is today. Later when we introduce different scenarios this parameter π will be changed.

Each production site $p \in P$ has an associated production cost given by $Prodcost^p$, reflecting the cost of manufacturing one unit of green ammonia.

Together, these parameters describe the economic structure of the network and provide the data necessary to evaluate transportation, production, and shortage decisions within the model.

Decision Variables:

Furthermore we introduce several decision variables that describe flow, production level and demand fulfilment. For each production site $p \in P$ and for every edge (i, j) the variable $f_{i,j}^{p}$ denotes the quantity of ammonia produced at production site p shipped along the edge going from node i to node j. This variable ensures that it is possible to track where the green ammonia delivered to off take center $o \in O$ is produced. The variable $prod^{p}$ represents the total production of ammonia on production site $p \in P$. At the off take centers, the variable q^{o} specifies the amount of ammonia delivered to off take center $o \in O$, while u^{o} is the unmet demand at the off take center. All variables are constrained to be nonnegative. Together, these decision variables capture the essential operational choices of the system, enabling the model to determine optimal production levels, routing decisions, and the extent to which demand is met across the network.

The model:
We end up with the following model

$$\min \quad \sum_{p\in P}\sum_{i\in N}\sum_{j\in N} c_{i,j}\, f_{i,j}^{p} \;+\; \text{Prodcost}^{p}\,\text{prod}^{p} \;+\; \sum_{o\in O} \pi\, u^{o} \tag{4.1}$$

$$\text{s.t.} \quad \sum_{j\in N} f_{p,j}^{p} = \text{prod}^{p} \qquad \forall p \in P \tag{4.2}$$

$$\text{prod}^{p} \leq \text{MaxP}^{p} \qquad \forall p \in P \tag{4.3}$$

$$\sum_{p\in P}\sum_{i\in N} f_{i,o}^{p} = q^{o} \qquad \forall o \in O \tag{4.4}$$

$$q^{o} + u^{o} \geq D^{o} \qquad \forall o \in O \tag{4.5}$$

$$\sum_{j\in N} f_{t,j}^{p} = \sum_{i\in N} f_{i,t}^{p} \qquad \forall p \in P, \forall t \in T \tag{4.6}$$

$$f_{i,j}^{p} = 0 \qquad \forall p \in P, \forall i \in P\backslash\{p\} : i \neq p, \forall j \in N \tag{4.7}$$

$$f_{i,j}^{p} \geq 0, \quad q^{o} \geq 0, \quad u^{o} \geq 0, \quad \text{prod}^{p} \geq 0 \tag{4.8}$$

Objective:
The objective function represents the total economic cost of operating the network and is minimised to identify an efficient allocation of production and transportation resources. The objective function captures all cost elements that contribute to the marginal cost of delivering one unit of ammonia to each off take center node $o \in O$. Transportation costs are included through the edge specific coefficients $c_{i,j}$, ensuring that the model accounts for the logistical expense of moving green ammonia from production sites through transfer points and onward to off take centers. Production costs at each production site are incorporated through $Prodcost_{p}$, reflecting the cost of producing green ammonia at the production sites. Finally, a penalty cost is assigned to unmet demand. This term is essential because the total available supply of green ammonia is insufficient to satisfy all demand in the system. By incorporating unmet demand in the objective, the model explicitly recognises the scarcity of green ammonia and internalises the economic consequences of that scarcity.

Constraints:
First, production capacity at each production site is limited in the constraint 4.3. Together

with constraint 4.2, it ensures that the amount of green ammonia shipped from each production site $p \in P$ do not exceed the maximum amount of green ammonia that each production site can produce. Constraint 4.6 ensures flow conservation in each transit point in the model. This flow conservation is based on where the green ammonia that is being transported is produced. Hence the incoming flow of ammonia produced at production site $p \in P$ is equal to the outgoing flow of ammonia produced at the same production site. This constraint takes all transit points $t \in T$ into consideration. Two constraints specify how delivered quantities and unmet demand are accounted for at the off take centers. Constraint 4.4 defines the delivered quantity q_o as the total flow arriving at each off take center node from all upstream nodes. Constraint 4.5 enforces that the sum of delivered and unmet demand must match the total demand at each off take center $o \in O$, ensuring accurate allocation of the produced green ammonia. Constraint 4.7 force the flow of ammonia from production site $p \in P$ on edges originating from any other production site to be zero. This ensure that green ammonia produced at production site $p \in P$ is only shipped from this production site. Lastly, all decision variables are continuous and nonnegative set in constraint 4.8.

Together, these constraints ensure that the model reflects the physical realities of production, transportation, and demand fulfillment in the network.

5 Input data processing

Several datasets were used to construct the model inputs, including information received from MMCZCS as well as publicly available data sources. These were constructed under different assumptions in the same way as the model. The data consists of transportation costs, ammonia production locations and estimated ammonia consumption locations derived from industrial production and fuel needs.

5.1 Modeling choises

Describe willingness to pay and why we are not ging that way. And also how it will still be incoorporated in a way with the fixed demand based on the different goals for green ammonia.

5.2 Graph creation

5.3 Demand

5.3.1 Shipping industry

Annual global marine fuel supply / bunkering volume (million metric tonnes per year) (i.e. how much bunker fuel can realistically be delivered worldwide) Bunkering marked is also growing 2026 = 249.65 Million tonnes fuel oil [Bunker fuel marked size and growth] The Top□10 bunkering ports alone supplied 111.9 Mt in 2024, representing approx 45 percent of global volume. Singapore by itself supplied 54.9 Mt in 2024, 22 percent of global bunkering [global top 10 bunkering]

We find that to transfer too the same energy unit we look at the LHV (Lower heating value) of ammonia and marine fuel oil. We found that fuel oil used now is a mix of diffent oils. As an approximation of their LHV we chose 41 MJ / kg. We found that the LHV for Ammonia is 18,6 MJ / Kg. [Brug samme kilde med road map IEA]. This gives

$$\frac{18.6}{41} \approx 2.2 \text{ kg } NH_3 \text{ per kg fuel oil}$$

Hence approx 2.2 tonnes of ammonia deliver the same energy as 1 tonne of marine fuel oil.

The total world wide demand of ammonia in 2026 is then:

$$2.2 \cdot 249.65 \text{million tonnes} = 549.23 \text{million tonnes}$$

5.3.2 Fertilizer

5.3.3 Steel

5.4 Ammonia production

The ammonia production levels used in the model are based on forecasts of future green ammonia production. Current global ammonia production is predominantly based on fossil fuels (grey ammonia); therefore, projections of future green ammonia capacity were used to represent potential production in a decarbonised energy system. **INSERT SOURCE on the green ammonia production forecasting**.

5.4.1 Unmet ammonia demand

Unmet ammonia demand is included in the model to ensure feasibility in cases where projected green ammonia production is insufficient to satisfy total demand. A penalty cost is assigned to each unit of unmet demand in the objective function.

The penalty is set to a sufficiently high value to discourage unmet demand whenever ammonia can be supplied within the modeled system. However, it allows the optimisation problem to remain feasible when supply constraints prevent the full demand from being satisfied.

5.5 Transportation

5.5.1 Routes and distances

To calculate routes and distances between the production sites, transit sites, and off-take centers, regional information for both production ports and offtake ports was provided. Based on this information, geographical coordinates were assigned to each port and stored in a CSV file containing the location name, identifier, and map coordinates. The model allows shipping between all ports, subject to the constraint that truck transport is limited to a maximum distance of 400 km.

Distances for maritime transport were calculated using the searoute Python package, which estimates realistic sea routes between ports. Road transport routes and distances were calculated using the openrouteservice Python API [INSERT SOURCE ON OPENROUTESERVICE AND SEAROUTE]. This is computationally heavy as it loads all maps in the chosen areas and tracks a route, so we decided to create a function that uses the coordinates as input and calculates the distances and the coordinates used in the route. Then this is saved for later use in the model.

5.5.2 New approach to onshore transportation of green ammonia

Early development of the model included the use of OpenRouteService to obtain realistic truck routes for the onshore transportation between transit nodes and production sites or off-take centers, mirroring the way sailing route distances are calculated. However, as the model is expanded to include multiple transport modes, such as rail, a unified method for calculating onshore transportation distances becomes necessary to ensure consistency. For this purpose, the Haversine function has been chosen. The Haversine formula is a standard and widely used method for computing great-circle distances between two points on the Earth's surface based solely on their latitude and longitude.

The distance is calculated by determining the central angle between two points on a sphere and converting this angle into a linear distance. This makes the method suitable for applications where network-specific route data are either unavailable or not comparable across transport types. This distance-estimation approach is also used in other studies where green ammonia or similar energy carriers are transported onshore [Insert reference and/or more description], demonstrating that spherical distance calculations provide a robust and consistent basis for cost modelling across different transport modes.

The Haversine formula is given by:

$$d = 2r \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\varphi}{2}\right) + \cos(\varphi_1)\cos(\varphi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)}\right)$$

where d is the distance between the two points, r is the Earth's radius, φ_1 and φ_2 are the latitudes of the points, $\Delta\varphi$ is the difference in latitude, and $\Delta\lambda$ is the difference in longitude.

Within this model, the formula is used to generate consistent “as-the-crow-flies” distance estimates for all onshore transportation modes. These distances are then applied uniformly in the cost calculations, ensuring that the comparison between truck, rail, and other potential modes is not affected by differences in network structure or data availability. [Find source]

We base our analysis on the transportation cost estimates for ammonia from 2019 [Indsæt reference (kun Olivia kan opdatere Mendelay]. Transportation cost data are based on several transportation modes and their respective price levels. Since our model requires a single aggregated cost to be multiplied by the Haversine distance, we compute the mean cost across all available transport modes and use this average as our baseline value. The resulting trend is illustrated in Figure 5.1.

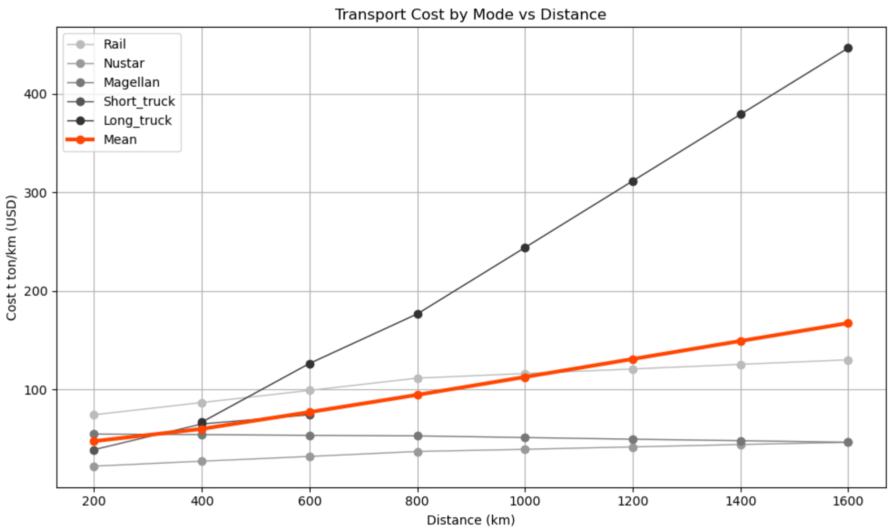

Figure 5.1: Transportation cost for ammonia in 2019

The full dataset underlying this calculation is provided in the appendix; see B.

5.6 Creation of $c_{i,j}$

5.6.1 Costs

The shipping costs for ammonia transport are based on analyses conducted by the MMCZSC. These estimates were used to represent maritime transport costs between nodes in the model. The dataset provides estimated shipping costs between regions and forms the basis for calculating ammonia transport costs by ship within the model.

The cost of transporting ammonia by truck was derived from publicly available data sources. A maximum transport radius of 400 km was assumed for road transport, as transporting ammonia over longer distances by truck would generally be economically inefficient compared to maritime transport or other alternatives [INSERT SOURCE ON TRUCK DISTANCE AND COST].

************ NEW ASSUMPTIONS *************

5.7 Flexible shipping demand

Unlike fertiliser and steel buyers whose demand is geographically fixed at plant locations, shipping demand is modeled as locationally flexible. Vessel operators choose bunkering ports endogenously based on delivered ammonia prices, consistent with observed bunkering optimisation behavior in practice. Total fleet demand D^{ship} must be met across the metwork, but the model can choose the optimal locations based on cheapest cost. This will align well with two scenarios that has production that will exceed consumption and one that will not meet the global demand. It can reveal optimal utilisation of the production regions and reveal whcih ports are optimal.

5.8 Shipping nodes

Transit ports are selected from the NGA World Port Index [9], a comprehensive global dataset published by the National Geospatial-Intelligence Agency. The WPI classifies each port into one of four harbor size categories, Large, Medium, Small, and Very Small, based on a composite assessment of harbor area, available facilities, and wharf space [9]. We retain only ports classified as Large or Medium, and additionally require the presence of liquid bulk handling facilities (harbor_size $\in$ {Large, Medium} and fac_liquid = Yes). This dual criterion ensures that retained transit nodes have both the physical scale and the specific infrastructure required to handle ammonia shipments at commercially relevant volumes. Small and Very Small ports are excluded on the grounds that their limited facilities make them unsuitable as transit nodes for large-scale ammonia trade, and their inclusion would not materially alter optimal routing given that larger ports provide regional coverage across all major maritime corridors. This selection yields a set of T transit nodes and improves the computational tractability of the global network model without sacrificing representational accuracy.

In addition to serving as transit nodes, all retained ports are also designated as potential off-take centers for shipping fuel demand, yielding the set O^{ship} This reflects the locationally flexible nature of maritime bunkering, vessel operators choose fueling locations based on delivered price rather than being tied to a fixed destination. The model therefore treats every retained port as a candidate bunkering node, and endogenously determines which ports supply ammonia to the shipping sector through the aggregate demand constraint and the flow optimisation.

5.9 Remaining sectors

Rather than modeling individual plants or facilities, demand is represented at the level of geographic regions disaggregated by sector. For each region $r \in R$ and each sector $s \in S = \{\text{fert, chem, steel, ship}\}$, a dedicated demand node (r, s) is created if sector s has nonzero ammonia demand in region r. This yields the full set of off-take centers:

$$O = \{(r, s) \ : \ r \in R, \ s \in S, \ D_{r,s} > 0\} \tag{5.1}$$

Each node $(r, s) \in O$ carries the aggregated ammonia demand of sector s within region r, computed from the MMCZCS dataset as described below.

Unlike the steel and shipping sectors, fertiliser production is directly comparable to ammonia demand, as ammonia is already used as a primary feedstock in fertiliser manufacturing. The MMCZCS production levels for the fertiliser sector can therefore be used directly as ammonia demand estimates without requiring a conversion factor.

The fraction of ammonia demand relative to steel production is assumed to be 0.31, with the specific calculation provided in A.1.1. However, this factor is not applied to the total steel production. Instead, it is multiplied by the share of steel production assumed to transition to hydrogen-based DRI. This reflects that only a portion of global steel production is expected to adopt ammonia-based hydrogen supply, and the transition share is defined through the scenario assumptions described in chapter 6. Thus, ammonia demand from the steel sector depends on both the ammonia intensity of hydrogen-based steel production and the assumed share of steel production that transitions to this technology. A similar assumption applies to the shipping sector, where ammonia is not currently used as a fuel but may substitute heavy fuel oil (HFO) in the future. The fraction is based on the lower heating values of ammonia and HFO. The detailed calculation of the conversion factor used in the model is provided in A.1.2.

6 Scenarios

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A Calculations

A.1 Ammonia demand

A.1.1 Steel industry

Hydrogen-based steel production requires approximately 0.055 tons of hydrogen per ton of steel produced. [10] The hydrogen mass fraction in ammonia is [11]:

$$\frac{3}{17}$$

This corresponds to:

$$1 \text{ ton } NH_3 = 0.176 \text{ ton } H_2$$

The ammonia requirement is then

$$\frac{0.055}{0.176} \approx 0.31$$

Therefore, approximately a factor of 0.31 is used to convert the ammonia used in steel.

A.1.2 Shipping industry

The conversion of HFO to ammonia is reliant on the lower heating value [12]

$$\text{HFO} = 40.5 \text{ MJ/kg}$$

$$NH_3 = 18.6 \text{ MJ/kg}$$

$$1\text{t HFO} = 2.17\text{t } NH_3$$

B Transportation cost for ammonia

Distance (km)	Rail	Nustar Pipeline	Magellan Pipeline	Short-dist. truck	Long-dist. truck	Mean
200	74.40	22.32	54.87	39.06	–	47.6625
400	86.86	27.34	54.31	65.10	66.96	60.1152
600	99.14	32.18	53.57	74.40	126.48	77.1528
800	111.60	37.20	53.01	–	176.70	94.6275
1000	116.25	39.43	51.34	–	244.03	112.7625
1200	120.90	41.85	49.66	–	311.55	130.9905
1400	125.55	44.27	48.17	–	379.07	149.2650
1600	130.20	46.50	46.50	–	446.40	167.4000

Table B.1: Transport Costs by Mode and Distance (multiplied by 18.6)

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